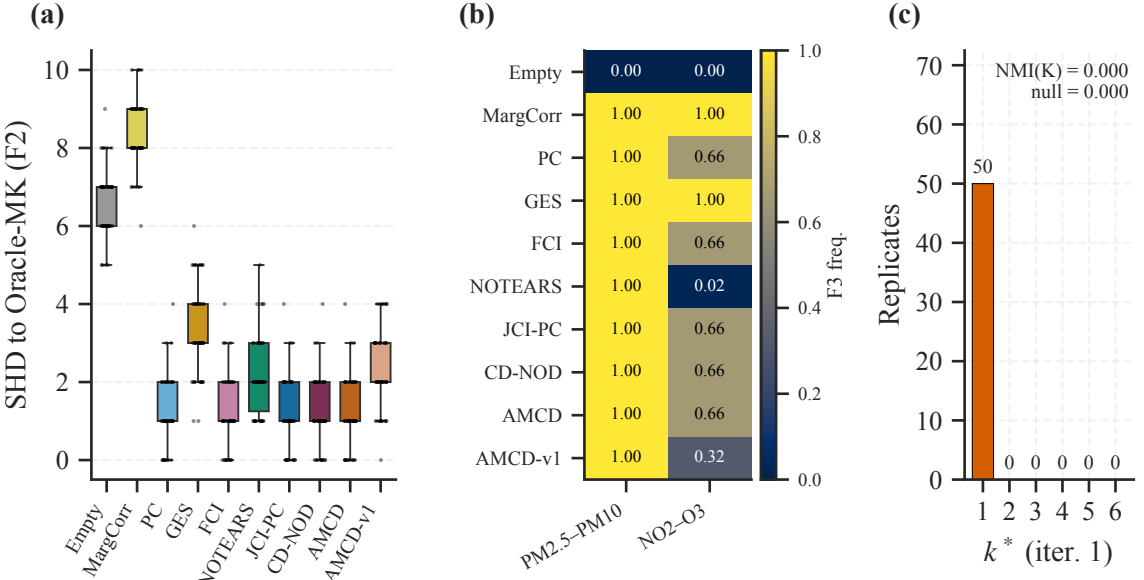

(a)
SHD to Oracle-MK (F2)
0
2
4
6
8
10
Empty
MargCorr
PC
GES
FCI
NOTEARS
JCI-PC
CD-NOD
AMCD
AMCD-v1
(b)
Empty 0.00 0.00
MargCorr 1.00 1.00
PC 1.00 0.66
GES 1.00 1.00
FCI 1.00 0.66
NOTEARS 1.00 0.02
JCI-PC 1.00 0.66
CD-NOD 1.00 0.66
AMCD 1.00 0.66
AMCD-v1 1.00 0.32
PM2.5–PM10
NO2–O3
F3 freq.
0.0
0.2
0.4
0.6
0.8
1.0
(c)
NMI(K) = 0.000
null = 0.000
Replicates
0
10
20
30
40
50
60
70
50
0 0 0 0 0
1 2 3 4 5 6
k * (iter. 1)